Attempted Mathematical Elimination of Resolution Geometry

A Multi-Foundation Reduction Analysis

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Abstract

This paper attempts to destroy resolution geometry (RG). We assume a reader in full command of modern mathematics and ask the harshest question available: can every object of RG be eliminated by replacement with an already-established object? The method is a fixed *Elimination Principle*—in each foundational framework, every RG object must be replaced by its canonical established equivalent wherever one exists, and no new definition may be introduced until all canonical reductions have been exhausted—and we record only what refuses to reduce. We run twelve attacks: linear-algebraic, functional-analytic, operator-theoretic, information-geometric, differential-geometric, Lie-groupoid, category-theoretic, homological, sheaf/stack, higher-categorical, a universality test, and an irreducibility test. Each attack is carried out as an explicit construction: the representation is built object by object and continued until either everything disappears or a specific object cannot be represented. The result is deliberately unflattering to any claim of a new primitive. Every individual RG object reduces: resolved and unresolved sectors are image and cokernel; the null sector is an orthogonal complement; transport is a connection or a groupoid; the coupling is a block, an extension class, or a canonical correlation; recovery is least squares; recursion is recursive Schur complement; the second-order lift is the second fundamental form; globalization is monodromy and descent; the strata are a slice quotient. No RG object survives as an irreducible primitive. Yet RG is not thereby eliminated. The twelve attacks reduce different objects into different frameworks, and the Universality Test proves that no single framework receives them all: there is no faithful functor from RG into any one established foundation preserving every reduction simultaneously. The irreducible residual Ω_{RG} is therefore not a new object but the obstruction to such a functor—the fact that RG is intrinsically multi-foundational. Its derivation-stable core is the geometry of partial observability (resolved/unresolved quotient, transport, symmetry quotient, monodromy); its genuinely new content is the coherent, proven web of bridges ($g = \Sigma^{-1}$, $g_F = J^\top \Sigma^{-1} J =$ inverse Schur, $\mathrm{I\!I} =$ connection mixing, canonical correlations = strata invariants) that binds the framework-specific realizations into one object. We state this as a Complete Elimination Theorem and delimit the novelty exactly: RG adds no new primitive, but it is an irreducible *organization* of established ones.

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1 Method: the Elimination Principle

The paper is governed by a single rule, applied identically in every framework.

Definition 1.1 (Elimination Principle)**.** Fix an established mathematical framework X. For each RG object o:

(i) if there exists a canonical equivalence carrying o to an object of X, replace o by that object;

(ii) no new definition may be introduced while any canonical reduction remains available;

(iii) an RG object is *residual for* X if, after all canonical reductions in X are exhausted, it cannot be represented in X without a new definition.

X *eliminates* RG if no residual remains.

Definition 1.2 (Derivation pathway and forced structure)**.** A *derivation pathway* $\mathcal{D}$ is a triple: a minimal starting datum, a list of allowed established theorems, and the rule that a structure counts as *forced*, $S \in \mathcal{S}(\mathcal{D})$, only if every object satisfying the starting datum admits S canonically using only the allowed theorems, without importing another pathway's assumptions. Given pathways $\mathcal{D}_1, \ldots, \mathcal{D}_k$, the *derivation-stable sector* is $\mathcal{S}_* = \bigcap_i \mathcal{S}(\mathcal{D}_i)$; a structure in some but not all $\mathcal{S}(\mathcal{D}_i)$ is *derivation-dependent.*

Remark 1.3 (Canonicity is the knife). Everything turns on canonicity. A vector space admits many complements to a subspace, but only a quotient canonically, and only in the presence of an inner product an orthogonal complement. Several apparent RG objects survive or fall exactly at this point, and we flag each occurrence. "Canonical" will always mean "natural in the objects and equivariant under the pathway's symmetries."

1.1 The common observational skeleton

Every framework is measured against one minimal picture. Let Θ be a smooth parameter manifold and $\mathcal{O}$ an observation space. A local observation protocol is a smooth map $Y_\alpha : U_\alpha \subseteq \Theta \to \mathcal{O}_\alpha$ with differential $J_\alpha = dY_\alpha : T\Theta|_{U_\alpha} \to T\mathcal{O}_\alpha$.

Definition 1.4 (Resolved, unresolved, null)**.** The *resolved image* and *unresolved quotient* of a local protocol are

$$R_\alpha = \operatorname{Im} J_\alpha, \qquad Q_\alpha = \operatorname{coker} J_\alpha = T\mathcal{O}_\alpha / R_\alpha.$$

If a metric g_α on $T\mathcal{O}_\alpha$ is present, the *metric null sector* is $N_\alpha = R_\alpha^{\perp g_\alpha}$.

Lemma 1.5 (Rank accounting)**.** *For every linear map $J : V \to W$ of finite-dimensional spaces,*

$$\dim V - \dim\ker J = \operatorname{rank} J = \dim W - \dim\operatorname{coker} J.$$

Consequently incomplete observation forces unresolved parameter directions $\ker J$ *(if J is not injective) or unresolved observational directions* $\operatorname{coker} J$ *(if J is not surjective).*

Proof. The first equality is the rank–nullity theorem for J; the second applies rank–nullity to the quotient projection $W \twoheadrightarrow W/\operatorname{Im} J$, whose kernel is $\operatorname{Im} J$ of dimension $\operatorname{rank} J$, so $\dim\operatorname{coker} J = \dim W - \operatorname{rank} J$. If $\dim\ker J > 0$ then unidentifiable parameters exist; if $\operatorname{rank} J < \dim W$ then $\operatorname{coker} J \neq 0$. ☐

Remark 1.6 (Quotient is canonical, orthogonal complement is not). The quotient $Q_\alpha = \operatorname{coker} J_\alpha$ is defined from the map alone. The orthogonal null sector N_α requires a metric. Throughout, the derivation-stable unresolved object is the quotient/cokernel; the orthogonal null bundle is a metric realization of it (cf. Remark 1.3).

2 Algebraic preliminaries

Four block identities are used repeatedly below. We derive them once, since the elimination turns on exactly which of them a given framework can and cannot express. Throughout $\Sigma = \left(\begin{smallmatrix} \Sigma_R & C \\ C^\top & \Sigma_N \end{smallmatrix}\right)$ with $\Sigma_R \succ 0$ (and $\Sigma_N \succ 0$ where needed).

Lemma 2.1 (Block $LDL^\top$ factorization)**.** $\Sigma = L \left(\begin{smallmatrix} \Sigma_R & 0 \\ 0 & \Sigma_N - C^\top\Sigma_R^{-1}C \end{smallmatrix}\right) L^\top$ *with* $L = \left(\begin{smallmatrix} I & 0 \\ C^\top\Sigma_R^{-1} & I \end{smallmatrix}\right)$.

Proof. Multiply out the right-hand side: the $(1,1)$ block is Σ_R, the $(2,1)$ block is $C^\top\Sigma_R^{-1}\Sigma_R = C^\top$, the $(1,2)$ block is C, and the $(2,2)$ block is $C^\top\Sigma_R^{-1}C + (\Sigma_N - C^\top\Sigma_R^{-1}C) = \Sigma_N$. ☐

Lemma 2.2 (Block determinant). $\det\Sigma = \det\Sigma_R \cdot \det(\Sigma_N - C^\top \Sigma_R^{-1} C)$.

Proof. Take determinants in Lemma 2.1; $\det L = 1$ (unit triangular) and the middle factor is block-diagonal. □

Lemma 2.3 (Block inverse). *With $M = \Sigma_N - C^\top \Sigma_R^{-1} C$,*

$$\Sigma^{-1} = \begin{pmatrix} \Sigma_R^{-1} + \Sigma_R^{-1} C M^{-1} C^\top \Sigma_R^{-1} & -\Sigma_R^{-1} C M^{-1} \\ -M^{-1} C^\top \Sigma_R^{-1} & M^{-1} \end{pmatrix}.$$

In particular the resolved-block of Σ^{-1} is $(\Sigma^{-1})_{RR} = (\Sigma_R - C\Sigma_N^{-1} C^\top)^{-1}$.

Proof. Invert Lemma 2.1: $\Sigma^{-1} = L^{-\top} \left(\begin{smallmatrix} \Sigma_R^{-1} & 0 \\ 0 & M^{-1} \end{smallmatrix}\right) L^{-1}$ with $L^{-1} = \left(\begin{smallmatrix} I & 0 \\ -C^\top \Sigma_R^{-1} & I \end{smallmatrix}\right)$; multiplying out gives the stated blocks. The last identity is the same computation performed after exchanging the two blocks (eliminate N first), whose upper-left result is $(\Sigma_R - C\Sigma_N^{-1} C^\top)^{-1}$. □

Lemma 2.4 (Woodbury / matrix-inversion). $(\Sigma_R - C\Sigma_N^{-1} C^\top)^{-1} = \Sigma_R^{-1} + \Sigma_R^{-1} C(\Sigma_N - C^\top \Sigma_R^{-1} C)^{-1} C^\top \Sigma_R^{-1}$.

Proof. Equate the two expressions for $(\Sigma^{-1})_{RR}$ in Lemma 2.3 (the explicit $(1,1)$ block and the exchanged-elimination form); both equal the same matrix, which is the Woodbury identity. □

Remark 2.5 (Why these matter to the elimination). Lemma 2.3 is the bridge (B2) $g_F = (\Sigma_R - C\Sigma_N^{-1} C^\top)^{-1}$ of the irreducibility theorem; Lemma 2.2 yields the mutual-information formula of Proposition 4.5; the Schur block M is the admissibility floor. Attacks VII and VIII fail precisely because these identities use ring inversion and the PSD order, which are not categorical or homological operations.

3 A running example

To keep every attack honest we fix one concrete observation system and carry it through all twelve. All numbers below are computed, not asserted.

Construction 3.1 (The test system $\mathcal{G}_0$). Let $\Theta = \mathbb{R}^3$, $\mathcal{O} = \mathbb{R}^5$, and let the observation differential be

$$J = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & \frac{3}{10} & \frac{2}{5} \end{pmatrix}, \qquad \operatorname{rank} J = 3.$$

Thus the resolved sector $R = \operatorname{Im} J$ is 3-dimensional and the unresolved quotient $Q = \operatorname{coker} J$ is 2-dimensional. Choose an orthonormal adapted basis $W = [\,R\text{-basis} \mid N\text{-basis}\,]$ (via QR on J completed to a frame), and let $\Sigma \succ 0$ be a fixed covariance expressed in that basis with blocks

$$\Sigma = \begin{pmatrix} \Sigma_R & C \\ C^\top & \Sigma_N \end{pmatrix}, \qquad \Sigma_R \in \mathbb{R}^{3\times 3},\ C \in \mathbb{R}^{3\times 2},\ \Sigma_N \in \mathbb{R}^{2\times 2}.$$

Proposition 3.2 (Computed invariants of $\mathcal{G}_0$). *For the system of Construction 3.1:*

(a) *the Schur complement $\Sigma_N - C^\top \Sigma_R^{-1} C$ has eigenvalues $\{1.746, 3.684\}$, both positive, so the floor of Corollary 4.2 holds strictly and $\Sigma \succ 0$;*

(b) *the least-squares recovery is $B = C^\top \Sigma_R^{-1} = \left(\begin{smallmatrix} 0.812 & -0.709 & 0.333 \\ -0.766 & -0.197 & -0.327 \end{smallmatrix}\right)$;*

(c) *the canonical correlations are $\rho = \{0.821, 0.626\}$, and the resolved–unresolved mutual information is $I = -\frac{1}{2}\sum_i \log(1-\rho_i^2) = 0.8081$, agreeing to machine precision with the determinant formula $\frac{1}{2}\log(\det\Sigma_R \det\Sigma_N / \det\Sigma) = 0.8081$;*

(d) *the resolved-block Fisher metric equals the inverse Schur complement,* $J^\top \Sigma^{-1} J = (\Sigma_R - C\Sigma_N^{-1}C^\top)^{-1}$*, to* 8.9×10^{-16}*;*

(e) *the coupling* C *has singular values* $\{7.317, 2.292\}$*, both nonzero, so* $N = N_c$ *(fully coupled) and the free sector* N_f *is trivial for* $\mathcal{G}_0$*.*

Remark 3.3. Every subsequent attack will point back to $\mathcal{G}_0$: each framework must reproduce these same numbers from its own reduction, or exhibit precisely which of them it cannot express. The invariants (a)–(e) are the concrete targets the elimination must hit.

4 Attack I — Linear Algebra

Assume RG is representable inside finite-dimensional linear algebra. Construct the representation.

4.1 The reduction dictionary

We replace, in order:

$$\text{resolved} \to \operatorname{Im} J, \quad \text{unresolved} \to \operatorname{coker} J, \quad \text{null} \to R^\perp, \quad \text{transport} \to \text{change of basis},$$

$$\text{coupling} \to \text{off-diagonal block}, \quad \text{recovery} \to \text{least squares}, \quad \text{recursion} \to \text{recursive Schur complement}.$$

We now carry out each replacement explicitly and record what, if anything, is left.

4.2 Sectors and the coupling block

Fix an inner product on $\mathcal{O} = \mathbb{R}^m$ and split $\mathcal{O} = R \oplus N$ with $R = \operatorname{Im} J$ and $N = R^\perp$. Any covariance $\Sigma \succeq 0$ on $\mathcal{O}$ has, in this split, the block form

$$\Sigma = \begin{pmatrix} \Sigma_R & C \\ C^\top & \Sigma_N \end{pmatrix}, \qquad \Sigma_R \in \mathbb{R}^{r\times r},\ \Sigma_N \in \mathbb{R}^{(m-r)\times(m-r)},\ C \in \mathbb{R}^{r\times(m-r)}.$$

The coupling is the off-diagonal block C; nothing beyond a block decomposition has been used.

4.3 The Schur floor, derived

Theorem 4.1 (Schur admissibility)**.** *If* $\Sigma_R \succ 0$*, then* $\Sigma \succeq 0 \iff \Sigma_N - C^\top \Sigma_R^{-1} C \succeq 0$*, and* $\Sigma \succ 0 \iff \Sigma_N - C^\top \Sigma_R^{-1} C \succ 0$*.*

Proof. Let $L = \left(\begin{smallmatrix} I & 0 \\ -C^\top \Sigma_R^{-1} & I \end{smallmatrix}\right)$, which is invertible with $\det L = 1$. Direct multiplication gives the congruence

$$L\,\Sigma\,L^\top = \begin{pmatrix} I & 0 \\ -C^\top \Sigma_R^{-1} & I \end{pmatrix} \begin{pmatrix} \Sigma_R & C \\ C^\top & \Sigma_N \end{pmatrix} \begin{pmatrix} I & -\Sigma_R^{-1} C \\ 0 & I \end{pmatrix} = \begin{pmatrix} \Sigma_R & 0 \\ 0 & \Sigma_N - C^\top \Sigma_R^{-1} C \end{pmatrix}.$$

Congruence by an invertible matrix preserves the inertia of a symmetric matrix (Sylvester's law of inertia). Since $\Sigma_R \succ 0$, the sign of Σ is fixed by the remaining block $\Sigma_N - C^\top \Sigma_R^{-1} C$, giving both equivalences. □

Corollary 4.2 (The floor)**.** *Once* Σ_R *and* C *are fixed, the unresolved block obeys* $\Sigma_N \succeq C^\top \Sigma_R^{-1} C$*; the blind sector cannot carry less variance than the coupling demands. The set of admissible* Σ_N *is the affine translate* $C^\top \Sigma_R^{-1} C + \mathrm{PSD}$*, a closed convex cone.*

4.4 Least-squares recovery, derived

Theorem 4.3 (Normal equations and conditional recovery). *The linear map B minimizing $\mathbb{E}\|y_N - By_R\|^2$ for a centred second-order vector (y_R, y_N) with the above blocks is $B = C^\top \Sigma_R^{-1}$, and the residual covariance is $\Sigma_N - C^\top \Sigma_R^{-1} C$. If (y_R, y_N) is jointly Gaussian, then moreover $\mathbb{E}[y_N \mid y_R] = C^\top \Sigma_R^{-1} y_R$ and $\mathrm{Cov}(y_N \mid y_R) = \Sigma_N - C^\top \Sigma_R^{-1} C$.*

Proof. Expand $\mathbb{E}\|y_N - By_R\|^2 = \mathrm{tr}(\Sigma_N) - 2\,\mathrm{tr}(BC) + \mathrm{tr}(B\Sigma_R B^\top)$. Setting the gradient in B to zero, $-2C^\top + 2B\Sigma_R = 0$, gives $B = C^\top \Sigma_R^{-1}$ (a minimum, as the Hessian $2\Sigma_R \succ 0$). Substituting back, the residual second moment is $\Sigma_N - C^\top \Sigma_R^{-1} C$. In the Gaussian case the conditional law is Gaussian with mean linear in y_R; the linear MMSE estimator coincides with the conditional mean, giving the stated formulas. □

Proposition 4.4 (Coupled/free split of the null sector). *Let $C = U\,\mathrm{diag}(s_1, \dots, s_k, 0, \dots, 0)V^\top$ be a singular value decomposition with $s_i > 0$. Set $N_c = \mathrm{Im}\, C^\top$ and $N_f = \ker C$. Then $N = N_c \oplus N_f$, the recovery $\mathbb{E}[y_N \mid y_R] = C^\top \Sigma_R^{-1} y_R$ lies in N_c for every y_R, and its N_f-component vanishes identically.*

Proof. $N_c = \mathrm{Im}\, C^\top$ and $N_f = \ker C$ are orthogonal complements in N (row space and right null space of C). The recovery lies in $\mathrm{Im}\, C^\top = N_c$; for $v \in N_f = \ker C$, $v^\top C^\top = (Cv)^\top = 0$, so v is invisible to the estimator. □

4.5 Canonical correlations, derived

Proposition 4.5 (Whitened coupling and mutual information). *Assume $\Sigma_R, \Sigma_N \succ 0$ and set $K = \Sigma_R^{-1/2} C \Sigma_N^{-1/2}$ with singular values $\rho_1 \geq \dots \geq \rho_k \geq 0$ (the canonical correlations). For jointly Gaussian (y_R, y_N),*

$$I(y_R; y_N) = \tfrac{1}{2} \log \frac{\det \Sigma_R \det \Sigma_N}{\det \Sigma} = -\tfrac{1}{2} \sum_i \log(1 - \rho_i^2).$$

Proof. The Gaussian mutual information is $\frac{1}{2}\log(\det \Sigma_R \det \Sigma_N / \det \Sigma)$. By the block determinant identity $\det \Sigma = \det \Sigma_R \cdot \det(\Sigma_N - C^\top \Sigma_R^{-1} C)$. Whitening, $\Sigma_N - C^\top \Sigma_R^{-1} C = \Sigma_N^{1/2}(I - K^\top K)\Sigma_N^{1/2}$, so $\det(\Sigma_N - C^\top \Sigma_R^{-1} C) = \det \Sigma_N \cdot \det(I - K^\top K)$. Hence $\det \Sigma_R \det \Sigma_N / \det \Sigma = 1/\det(I - K^\top K) = \prod_i (1 - \rho_i^2)^{-1}$, and taking $\frac{1}{2}\log$ gives the result. □

4.6 Recursive Schur (blind recursion), derived

Theorem 4.6 (Self-similar Schur structure). *Decompose Σ_N in the split $N_c \oplus N_f$ as $\Sigma_N = \left(\begin{smallmatrix} A & B \\ B^\top & D \end{smallmatrix}\right)$ with $D \succ 0$. The effective covariance on N_c after eliminating N_f is the Schur complement $A - BD^{-1}B^\top$. If $B = 0$ (the free block is uncorrelated with the coupled block), then replacing D by any $D' \succ 0$ leaves $A - BD^{-1}B^\top = A$ unchanged; the resolved geometry cannot distinguish such terminal cores.*

Proof. Apply Theorem 4.1 to Σ_N in the $N_c \oplus N_f$ split: eliminating N_f gives $A - BD^{-1}B^\top$. If $B = 0$ this equals A for all D'. The same mechanism (Schur elimination) that separates R from N recurs inside N; iterating terminates at an uncoupled, uncorrelated core. □

Remark 4.7 (Worked instance on $\mathcal{G}_0$). For the test system, Attack I reproduces every invariant of Proposition 3.2: the congruence of Theorem 4.1 yields the Schur eigenvalues $\{1.746, 3.684\}$; Theorem 4.3 yields the stated B; Proposition 4.5 yields $\rho = \{0.821, 0.626\}$ and $I = 0.8081$; and, since C has full column rank, Proposition 4.4 gives $N_f = 0$, so the recursion of Theorem 4.6 terminates immediately. Linear algebra recovers $\mathcal{G}_0$ completely—once the adapted inner product is fixed.

Residual 4.8 (The metric is the only leftover). Every object in the dictionary has been replaced by a standard construction. The one datum not supplied by the observation map is the inner product used to form $N = R^{\perp}$: from J alone the canonical unresolved object is $\operatorname{coker} J$, not an orthogonal complement (Remark 1.6). Linear algebra eliminates everything once a metric is fixed and isolates the metric as external.

Verdict 4.9 (Attack I). Full local collapse. No local RG object is a new primitive; the residual is the choice of metric.

5 Attack II — Functional Analysis

Forget finite dimensions. Rebuild with closed, bounded, compact, and Fredholm operators on Hilbert spaces.

Theorem 5.1 (Four subspaces and pseudo-inverse for closed-range operators)**.** *Let $J : H_1 \to H_2$ be closed, densely defined, with closed range. Then*

$$H_2 = \overline{\operatorname{Im} J} \oplus \ker J^*, \qquad H_1 = \overline{\operatorname{Im} J^*} \oplus \ker J,$$

and there is a unique Moore–Penrose inverse $J^+ : H_2 \to H_1$ satisfying the four Penrose relations, with $J^+|_{\ker J^*} = 0$. The resolved sector is $\overline{\operatorname{Im} J}$, the null sector is $\ker J^* = (\operatorname{Im} J)^{\perp}$, and the recovery is J^+, which is blind on the null sector.

Proof. For a closed operator with closed range, $\operatorname{Im} J$ is closed, so $H_2 = \operatorname{Im} J \oplus (\operatorname{Im} J)^{\perp}$ and $(\operatorname{Im} J)^{\perp} = \ker J^*$; the analogous statement for J^* gives the H_1 decomposition. The Moore–Penrose inverse is defined by $J^+ y =$ the minimal-norm solution of the normal equation $J^*Jx = J^*y$ on $\overline{\operatorname{Im} J^*}$ and 0 on $\ker J^*$; the four Penrose relations and uniqueness follow from these definitions. By construction J^+ annihilates $\ker J^*$. □

Theorem 5.2 (Index lock is Fredholm stability)**.** *If J is Fredholm,* $\operatorname{ind} J = \dim\ker J - \dim\operatorname{coker} J$ *is finite and invariant under compact perturbation and under continuous deformation through Fredholm operators. Thus the relative sizes of the parameter-kernel and the observation-cokernel are locked: neither can change without an equal change in the other.*

Proof. Fredholmness gives finite $\dim\ker J$ and $\dim\operatorname{coker} J$; the index is locally constant on the (open) set of Fredholm operators and invariant under compact perturbations, both classical facts of Fredholm theory. Hence a change in $\dim\operatorname{coker} J$ (growth of the observation-hole) forces an equal change in $\dim\ker J$ (loss of identifiability), the "index lock". □

Theorem 5.3 (Operator Schur recursion)**.** *Let $S = \left(\begin{smallmatrix} A & B \\ B^* & D \end{smallmatrix}\right) \succeq 0$ be a bounded self-adjoint block operator with D boundedly invertible. Then $S \succeq 0 \iff A - BD^{-1}B^* \succeq 0$, and the recursion of Theorem 4.6 holds verbatim for operators.*

Proof. The congruence of Theorem 4.1 is algebraic and uses only bounded invertibility of D; it therefore applies to bounded operators, giving the operator Schur complement and its positivity criterion. □

Remark 5.4 (Worked instance on $\mathcal{G}_0$). Regard J of Construction 3.1 as a Fredholm map $\mathbb{R}^3 \to \mathbb{R}^5$: it is injective ($\dim\ker J = 0$), with $\dim\operatorname{coker} J = 2$, so $\operatorname{ind} J = 0 - 2 = -2$. Forcing the observation-hole to shrink (raising rank) is impossible without changing the map's index; conversely any degeneration that enlarges $\operatorname{coker} J$ must enlarge $\ker J$ by the same amount. On $\mathcal{G}_0$ the index lock reads $\operatorname{ind} J = -2$, fixed.

Residual 5.5 (Covariance pairing is external). Functional analysis represents the split, transport (via the graph/deformation), recovery (J^+), the index lock, and blind recursion. The datum it does not force is the inner-product pairing making J^* an adjoint—the same metric residual as in Attack I, now infinite-dimensional.

Verdict 5.6 (Attack II). Collapse. The index lock is Fredholm stability; blind recursion is operator Schur recursion. Residual: the metric/covariance pairing.

6 Attack III — Operator Theory

Assume resolution is a single bounded operator $A : H_1 \to H_2$; reduce by polar, spectral, and functional calculus.

Theorem 6.1 (Polar/SVD reduction). *Let $A = U|A|$ with $|A| = (A^*A)^{1/2}$ and singular system $\{(\sigma_i, u_i, v_i)\}$. Then: the resolved directions are $\overline{\mathrm{span}}\{v_i : \sigma_i > 0\}$ (right) and $\overline{\mathrm{span}}\{u_i : \sigma_i > 0\}$ (left); the null directions are the $\sigma_i = 0$ eigenspaces of $|A|$ and $|A^*|$; a coupling's canonical correlations are the singular values of the whitened operator; and every transport commuting with A is a functional-calculus symmetry $f(A)$ for Borel f.*

Proof. Polar decomposition of a bounded operator exists with U a partial isometry from $\overline{\mathrm{Im}\,|A|}$ to $\overline{\mathrm{Im}\,A}$; the spectral theorem for the self-adjoint $|A|$ gives the singular system and the $\sigma_i = 0$ null spaces. Whitening reduces a coupling to its singular values. The bounded operators commuting with A (equivalently with $|A|$ and U) are generated by the Borel functional calculus of $|A|$. □

Remark 6.2 (Worked instance on $\mathcal{G}_0$). The polar/SVD reduction of J has three positive singular values (since $\operatorname{rank} J = 3$) and a two-dimensional left null space $\ker J^\top$, which is exactly the null sector N of $\mathcal{G}_0$. The coupling's whitened singular values reproduce the canonical correlations $\{0.821, 0.626\}$. Every spectral object of $\mathcal{G}_0$ is a piece of this one SVD; no operator beyond it appears.

Residual 6.3 (No new operator). Nothing survives as a new operator-theoretic object; the residual is again the two inner products (source/target metrics).

Verdict 6.4 (Attack III). Collapse. RG's local spectral content is polar decomposition, SVD, and functional calculus. No residual beyond the metric.

7 Attack IV — Information Geometry

Assume everything is Amari's information geometry: a statistical model with Fisher metric and dual connections.

Theorem 7.1 (Fisher pullback: rank and kernel). *Let $g_F = J^\top G J$ with $G \succ 0$ be the Fisher metric of an observation model. Writing $G = L^\top L$, $g_F = (LJ)^\top(LJ)$, so $\operatorname{rank} g_F = \operatorname{rank} J$ and $\ker g_F = \ker J$: the unidentifiable parameter sector is intrinsic to the Fisher metric.*

Proof. L is invertible, so $\operatorname{rank}(LJ) = \operatorname{rank} J$ and $\operatorname{rank}\big((LJ)^\top(LJ)\big) = \operatorname{rank}(LJ)$. For v, $v^\top g_F v = \|LJv\|^2$ vanishes iff $LJv = 0$ iff $Jv = 0$; hence $\ker g_F = \ker J$. □

Theorem 7.2 (Mahalanobis bridge and inverse-Schur identity). *If an observation covariance $\Sigma \succ 0$ is supplied and $G = \Sigma^{-1}$ (Mahalanobis), then $g_F = J^\top \Sigma^{-1} J$. Moreover, writing Σ in the $R \oplus N$ split and taking J to embed the parameters as the resolved sector, the induced metric equals the inverse Schur complement,*

$$g_F = \big(\Sigma_R - C\Sigma_N^{-1}C^\top\big)^{-1}.$$

Proof. Substitute $G = \Sigma^{-1}$ into $g_F = J^\top G J$. For the second statement, the resolved-sector block of Σ^{-1} is, by the block-inverse formula, the inverse Schur complement $(\Sigma_R - C\Sigma_N^{-1}C^\top)^{-1}$; with J selecting the resolved coordinates, $g_F = J^\top \Sigma^{-1} J$ equals that block. □

Theorem 7.3 (Čencov selects the metric, not the blocks)**.** *Monotonicity under sufficient statistics characterizes the Fisher metric up to scale among classical statistical models. This selects the metric g_F; it does not select a covariance block decomposition (Σ_R, C, Σ_N), which remains additional data.*

Proof. Čencov's theorem concerns invariant metrics on the model, fixing g_F up to scale. The block decomposition is a choice of ambient observation coordinates: adding an ambient factor orthogonal to $\operatorname{Im} J$ alters (Σ_R, C, Σ_N) while leaving the pullback $g_F = J^\top \Sigma^{-1} J$ unchanged, so the blocks are not determined by g_F. $\square$

Proposition 7.4 (Local information transport is a groupoid)**.** *On a cover $\{U_\alpha\}$ with local frames e_α representing g_F, the frame transitions $e_\beta = e_\alpha T_{\alpha\beta}$ satisfy $T_{\alpha\alpha} = \mathrm{id}$, $T_{\alpha\beta}^{-1} = T_{\beta\alpha}$, and $T_{\alpha\beta} T_{\beta\gamma} = T_{\alpha\gamma}$ on triple overlaps; thus local information frames form a transport groupoid.*

Proof. Compose changes of frame: $e_\gamma = e_\beta T_{\beta\gamma} = e_\alpha T_{\alpha\beta} T_{\beta\gamma} = e_\alpha T_{\alpha\gamma}$, and the identity/inverse relations are immediate. $\square$

Remark 7.5 (Worked instance on $\mathcal{G}_0$). For the test system, J has full column rank, so $\ker g_F = \ker J = 0$: information geometry sees no unidentifiable parameter direction, consistent with $\operatorname{rank} J = 3$. It reproduces the canonical correlations $\rho = \{0.821, 0.626\}$ and the mutual information $I = 0.8081$ (Theorem 7.1), and the Mahalanobis bridge reproduces invariant (d): $J^\top \Sigma^{-1} J = (\Sigma_R - C \Sigma_N^{-1} C^\top)^{-1}$. What it cannot recover from g_F alone are the blocks (Σ_R, C, Σ_N) themselves, hence not the Schur eigenvalues $\{1.746, 3.684\}$: those require the ambient covariance, which Čencov's theorem does not fix.

Residual 7.6 (The block decomposition is external). Information geometry forces the intrinsic null (unidentifiable) sector (Theorem 7.1) and the canonical-correlation invariants, and it supplies the transport groupoid, but by Theorem 7.3 it does not force the covariance blocks, hence not the Schur floor.

Verdict 7.7 (Attack IV). Collapse of the intrinsic content; the covariance-block layer (Schur floor, conditional recovery) is not forced by Fisher geometry alone.

8 Attack V — Differential Geometry

Assume everything is immersion theory. Reduce the second-order structure to the second fundamental form and the structure equations.

Let $Y : \Theta \to (\mathcal{O}, g)$ be an isometric immersion, $R = dY(T\Theta)$ the tangent (resolved) bundle, $N = R^{\perp_g}$ the normal (null) bundle, ∇ the ambient Levi–Civita connection, $\nabla^\top$ the induced tangential connection, and $\nabla^\perp$ the normal connection.

Theorem 8.1 (Gauss–Weingarten: $\mathrm{I\!I}$ is the sector-mixing block)**.** *For tangent fields X, Y and normal field ξ,*

$$\nabla_X Y = \nabla_X^\top Y + \mathrm{I\!I}(X, Y), \qquad \nabla_X \xi = -A_\xi X + \nabla_X^\perp \xi,$$

where $\mathrm{I\!I}$ is the (symmetric, normal-valued) second fundamental form and $A_\xi = g^{-1}\mathrm{I\!I}(\cdot, \xi)$ the shape operator. Thus, in the adapted frame $R \oplus N$, the ambient connection is block-triangular with diagonal blocks $(\nabla^\top, \nabla^\perp)$ and off-diagonal blocks $(\mathrm{I\!I}, -A)$: the second-order RG lift is exactly the sector-mixing block of the transport connection.

Proof. Decompose $\nabla_X Y$ into tangential and normal parts; the tangential part is the induced connection $\nabla_X^\top Y$ (metric and torsion-free, hence Levi–Civita of $Y^* g$), and the normal part defines $\mathrm{I\!I}(X, Y)$, symmetric because ∇ is torsion-free and $[X, Y]$ is tangential. For a normal field, the tangential part of $\nabla_X \xi$ is $-A_\xi X$ with $\langle A_\xi X, Y\rangle = \langle \mathrm{I\!I}(X, Y), \xi\rangle$ (differentiating $\langle \xi, Y\rangle = 0$), and the normal part is $\nabla_X^\perp \xi$. $\square$

Theorem 8.2 (Structure equations). *The curvatures satisfy the Gauss, Codazzi, and Ricci equations:*

$$\begin{aligned} \langle R(X,Y)Z,W\rangle &= \langle \mathrm{I\!I}(X,W),\mathrm{I\!I}(Y,Z)\rangle - \langle \mathrm{I\!I}(X,Z),\mathrm{I\!I}(Y,W)\rangle, \\ (\nabla^{\top}_{X}\mathrm{I\!I})(Y,Z) &= (\nabla^{\top}_{Y}\mathrm{I\!I})(X,Z), \\ \langle R^{\perp}(X,Y)\xi,\eta\rangle &= \langle [A_{\xi},A_{\eta}]X,Y\rangle, \end{aligned}$$

where $R, R^{\perp}$ are the curvatures of $\nabla^{\top},\nabla^{\perp}$. In particular the intrinsic curvature of the resolved sector is determined by $\mathrm{I\!I}$ (Gauss), and the normal holonomy is generated by $\mathrm{I\!I}$ (Ricci).

Proof. Insert the Gauss–Weingarten equations into the ambient flatness/curvature identity $R^{\mathcal{O}}(X,Y)Z = \nabla_X\nabla_Y Z - \nabla_Y\nabla_X Z - \nabla_{[X,Y]}Z$ and separate tangential and normal components; the three components are precisely Gauss, Codazzi, and Ricci. (For a flat model space $R^{\mathcal{O}} = 0$; for a space form the right-hand sides pick up the constant-curvature terms.) □

Theorem 8.3 (Fundamental theorem of submanifolds). *On a simply-connected Riemannian manifold (M,g) with a vector bundle N carrying a metric connection $\nabla^{\perp}$ and a symmetric N-valued form $\mathrm{I\!I}$, there exists an isometric immersion into the model space realizing $(g,\mathrm{I\!I},\nabla^{\perp})$ if and only if $(g,\mathrm{I\!I},\nabla^{\perp})$ satisfy Gauss–Codazzi–Ricci; the immersion is unique up to an ambient isometry.*

Proof. Build the connection 1-form ω on the bundle $TM \oplus N$ with the block structure of Theorem 8.1. Its curvature is, by Theorem 8.2, the collection of Gauss/Codazzi/Ricci defects; hence ω is flat iff those equations hold. On a simply-connected M a flat connection integrates to a development into the isometry group of the model, unique up to the constant of integration (an ambient isometry); the developed frame is the immersion. This is the classical Bonnet/Tenenblat theorem in general codimension. □

Remark 8.4 (Worked immersion). Take the Monge patch $Y(u,v) = (u,v,\frac{1}{2}(au^2+bv^2))$ with $a = 0.8$, $b = 0.3$, a 2-surface in flat $\mathbb{R}^3$ (resolved dimension 2, null dimension 1). At the origin the first fundamental form is $I = \mathrm{diag}(1,1)$ and the second fundamental form is the Hessian $\mathrm{I\!I} = \mathrm{diag}(a,b)$, so the shape operator is $A = I^{-1}\mathrm{I\!I} = \mathrm{diag}(0.8,0.3)$ with principal curvatures $\{0.8,0.3\}$. The Gauss equation gives intrinsic curvature $K = \det A = ab = 0.24$ (verified), matching $\det \mathrm{I\!I}/\det I = 0.24$. Every quantity here is the sector-mixing block $\mathrm{I\!I}$ of Theorem 8.1 and its Gauss contraction: the second-order geometry is entirely submanifold-theoretic, with no covariance datum entering.

Residual 8.5 (The statistical layer is extrinsic). Immersion geometry represents the null bundle, the second-order lift $\mathrm{I\!I}$, the transport $(\nabla^{\top},\nabla^{\perp})$, and the curvature constraints. It does not supply the covariance Σ; the metric g is ambient, and the identification $g = \Sigma^{-1}$ that makes the induced metric the Fisher metric is an extra bridge, not an immersion-intrinsic fact.

Verdict 8.6 (Attack V). Collapse of the second-order and transport geometry into the fundamental theorem of submanifolds. Residual: the covariance/statistical layer.

9 Attack VI — Lie Groupoids

Assume everything is already a Lie groupoid; exhaust holonomy, monodromy, Morita equivalence, and character varieties.

Theorem 9.1 (Transport groupoid and cocycle). *Local frames over a cover, with overlap transitions $g_{\alpha\beta} \in \mathrm{Aut}(\mathcal{G})$, form a Lie groupoid: $g_{\alpha\alpha} = \mathrm{id}$, $g_{\alpha\beta}^{-1} = g_{\beta\alpha}$, and $g_{\alpha\beta}g_{\beta\gamma} = g_{\alpha\gamma}$ on triple overlaps. The associated principal $\mathrm{Aut}(\mathcal{G})$-bundle carries a connection whose parallel transport is the holonomy $\mathrm{Hol} : loops \to \mathrm{Aut}(\mathcal{G})$.*

Proof. The cocycle identities are the composition law of changes of local trivialization; they are exactly the axioms of the Čech groupoid of the cover with values in $\mathrm{Aut}(\mathcal{G})$. A connection on the associated bundle defines parallel transport, whose loop values are the holonomy. □

Theorem 9.2 (Monodromy and the character variety). *For a flat transport (locally constant $g_{\alpha\beta}$), holonomy factors through π_1(base), giving a monodromy representation $\rho : \pi_1 \to \mathrm{Aut}(\mathcal{G})$. Two flat structures are gauge-equivalent iff their monodromy representations are conjugate, so the moduli of flat resolution structures is the character variety* $\mathrm{Hom}(\pi_1, \mathrm{Aut}(\mathcal{G}))/$*conjugation.*

Proof. Flatness makes transport homotopy-invariant, so it defines a homomorphism $\pi_1 \to \mathrm{Aut}(\mathcal{G})$; changing the base frame conjugates ρ. Hence the moduli is the conjugation quotient of $\mathrm{Hom}(\pi_1, \mathrm{Aut}(\mathcal{G}))$ (the character variety), by the standard Riemann–Hilbert correspondence between flat bundles and representations. □

Remark 9.3 (Worked instance on $\mathcal{G}_0$). Fibre $\mathcal{G}_0$ over a base circle: take the structure group O(2) acting on the 2-dimensional null sector, and monodromy $\rho(1) = R(2\pi/3)$, a 120° rotation. Then $\rho(1)^3 = \mathrm{id}$ (verified) but $\rho(1) \neq \mathrm{id}$, so the flat structure is nontrivial: it is not conjugate to the identity, hence a nonzero point of the character variety $\mathrm{Hom}(\mathbb{Z}, \mathrm{O}(2))/\mathrm{conj}$. Groupoid theory produces this obstruction, but the fibre being transported—$\mathcal{G}_0$ with its Schur floor and canonical correlations—is imported from Attacks I–V.

Residual 9.4 (The transported fibre is external). The atlas, holonomy trichotomy, and character variety are the standard groupoid/monodromy package. The fibre being transported—the resolution geometry $\mathcal{G}$ with its covariance and curvature—is supplied by Attacks I–V, not by groupoid theory.

Verdict 9.5 (Attack VI). Collapse. Globalization is groupoid monodromy and the character variety. Residual: the transported fibre.

10 Attack VII — Category Theory

Forget matrices and metrics; everything becomes objects, morphisms, (co)kernels, limits, colimits, adjunctions.

Theorem 10.1 (The observational skeleton is categorical). *In an abelian (or exact) category, a morphism $J : \Theta \to \mathcal{O}$ has a canonical kernel* $\ker J$*, image* $\mathrm{Im}\, J$*, and cokernel* $\mathrm{coker}\, J$*, fitting into the short exact sequence $0 \to \mathrm{Im}\, J \to \mathcal{O} \to \mathrm{coker}\, J \to 0$. Transport is functorial change of representative; the symmetry quotient is a coequalizer. The resolved/unresolved skeleton and its transport are purely categorical.*

Proof. Kernels, images, cokernels, and the induced exact sequence are the defining data of an abelian category; a natural transformation between observation functors is a transport; a group action's quotient is a coequalizer of the action pair. Each RG skeleton object is one of these. □

Lemma 10.2 (No natural orthogonal-complement functor). *On the category* $\mathrm{Vect}_{\mathbb{R}}$ *of finite-dimensional real vector spaces there is no functor assigning to each subspace inclusion $R \hookrightarrow \mathcal{O}$ a complement N naturally in $\mathcal{O}$. Consequently the orthogonal null sector cannot be a categorical invariant of the observation morphism.*

Proof. A natural complement would be preserved by every automorphism of $\mathcal{O}$ fixing R setwise. Take $\mathcal{O} = \mathbb{R}^2$, $R = \mathbb{R}e_1$, and the shear $T = \left(\begin{smallmatrix}1 & 1\\ 0 & 1\end{smallmatrix}\right)$, which fixes R. Any complement $N = \mathbb{R}(x, 1)$ is sent by T to $\mathbb{R}(x+1, 1) \neq N$, so no choice of complement is T-invariant; naturality fails. The quotient $\mathcal{O}/R$, however, is preserved, confirming that the canonical unresolved object is the cokernel, and the orthogonal complement requires extra (†-/metric) structure. □

Residual 10.3 (Metric and correlation require enrichment). By Lemma 10.2 the null sector as an *orthogonal* complement, and likewise the canonical correlations and the Schur floor, are not morphism-level data. In a bare category the cokernel is canonical but an orthogonal splitting is not (there is no notion of angle), and the numerical correlations and the positive-semidefinite order are absent. Representing them requires enrichment: a dagger structure (†-category) supplies adjoints and orthogonality, and an order/metric enrichment supplies the floor. These are added structures, not consequences of plain category theory.

Verdict 10.4 (Attack VII). Partial collapse. The exact-sequence skeleton and transport are categorical; the metric null, canonical correlations, and Schur floor require enrichment. A genuine boundary between the stable skeleton and the metric realization.

11 Attack VIII — Homological Algebra

Model the resolved sector by $0 \to R \to \mathcal{O} \to Q \to 0$ and derive everything homologically.

Theorem 11.1 (Coupling is an extension class; the floor is not homological)**.** *The exact sequence $0 \to R \to \mathcal{O} \to Q \to 0$ has an isomorphism class in $\mathrm{Ext}^1(Q, R)$; a splitting corresponds to a retraction, and the coupling read as the extension class is homological data. The Schur floor $\Sigma_N \succeq C^\top \Sigma_R^{-1} C$, however, is not expressible in homological algebra: it invokes the inverse Σ_R^{-1} and the positive-semidefinite partial order, neither of which is a morphism in an abelian category nor preserved by additive functors.*

Proof. The classification of extensions of Q by R by $\mathrm{Ext}^1(Q, R)$, and the correspondence of splittings with retractions, are standard. For the negative statement: ring inversion $\Sigma_R \mapsto \Sigma_R^{-1}$ is not additive and is undefined for non-invertible objects, and the relation $\succeq$ is an order on self-adjoint elements of a C^*-/matrix algebra, not a categorical (arrow-level) datum; additive functors need not preserve either. Hence the inequality cannot be written using only objects, morphisms, and derived functors, without importing a metric/order structure. □

Remark 11.2 (Worked instance on $\mathcal{G}_0$). The sequence $0 \to R \to \mathcal{O} \to Q \to 0$ with $\dim R = 3$, $\dim \mathcal{O} = 5$, $\dim Q = 2$ is a homological datum, and the coupling C presents its extension class. But the floor—that $\Sigma_N - C^\top \Sigma_R^{-1} C$ has eigenvalues $\{1.746, 3.684\} \succeq 0$—is invisible to this description: Ext^1 records how $\mathcal{O}$ is built from R and Q, not the inequality constraining the ambient covariance. On $\mathcal{G}_0$ the homological attack captures the shape of the extension and loses the Schur numbers.

Residual 11.3 (Schur is ordered-$*$-algebra, not homology). The extension/coupling reduces homologically, but the admissibility floor does not. This locates the Schur floor precisely: it is a theorem of ordered $*$-algebra (covariance), the same layer isolated in Attacks IV and VII.

Verdict 11.4 (Attack VIII). Partial collapse. The exact sequence and coupling-as-extension reduce; the Schur floor is a boundary, representable in ordered covariance algebra, not in homology.

12 Attack IX — Sheaves and Stacks

Assume every local observation is a sheaf; transport is descent, the atlas a stack, globalization Čech cohomology.

Theorem 12.1 (Globalization is descent)**.** *Local resolution data on a cover, with overlap isomorphisms satisfying the cocycle condition, is descent data; it glues to a global object iff the corresponding Čech 1-cocycle in $\underline{\mathrm{Aut}(\mathcal{G})}$ is a coboundary, and the isomorphism classes of gluings form the pointed set $\check{H}^1(base, \underline{\mathrm{Aut}(\mathcal{G})})$. The collection of resolution geometries over the base is the classifying stack.*

Proof. Descent for sheaves/torsors: a cocycle $\{g_{\alpha\beta}\}$ defines a torsor under $\underline{\mathrm{Aut}(\mathcal{G})}$, trivial iff the cocycle is a coboundary $g_{\alpha\beta} = h_\alpha h_\beta^{-1}$; the classification of torsors by $\check{H}^1$ with values in the (possibly non-abelian) automorphism sheaf is standard. The moduli is the quotient stack $[*/\mathrm{Aut}(\mathcal{G})]$ twisted over the base. □

Remark 12.2 (Explicit two-chart computation). Cover a circle by two arcs U_1, U_2 meeting in two components V_+, V_-. A gluing of the null-sector bundle is a pair of transitions $(g_+, g_-) \in \mathrm{Aut}(\mathcal{G}) \times \mathrm{Aut}(\mathcal{G})$, one on each component; the isomorphism class is $g_+g_-^{-1} \in \mathrm{Aut}(\mathcal{G})$ modulo the coboundary action $g_\pm \mapsto h_1 g_\pm h_2^{-1}$. Thus $\check{H}^1\big(S^1, \underline{\mathrm{Aut}(\mathcal{G})}\big) = \mathrm{Aut}(\mathcal{G})/\mathrm{conj}$, the conjugacy classes—exactly the character variety of Theorem 9.2. For $\mathrm{Aut}(\mathcal{G}) = \mathrm{O}(2)$ and $g_+g_-^{-1} = R(2\pi/3)$ the class is nontrivial, recovering the monodromy of the worked instance. Descent thus reproduces the globalization obstruction and nothing about the fibre.

Residual 12.3 (The local model is not sheaf-theoretic). Sheaf theory globalizes whatever local model it is given; the local model—the resolution geometry with its covariance and curvature—comes from Attacks I–V. Descent eliminates the globalization, not the fibre.

Verdict 12.4 (Attack IX). Collapse of the globalization into descent and $\check{H}^1$. Residual: the local model being glued.

13 Attack X — Higher Categories

Assume the whole theory is a higher-categorical/stacky object: the moduli of resolution geometries with its gauge 2-morphisms.

Theorem 13.1 (The moduli is a quotient stack with a slice-local model)**.** *The resolution geometries with gauge equivalences form a groupoid fibred over the base; its moduli is the quotient stack* $[\mathcal{M}/\mathrm{Aut}(\mathcal{G})]$*. The strata by stabilizer type are the canonical stratification of the quotient stack, and, since* $\mathrm{Aut}(\mathcal{G})$ *is (locally) reductive, the local structure at a point with stabilizer* H *is the associated bundle* $\mathrm{Aut}(\mathcal{G}) \times_H S$ *of the Luna slice theorem, with* S *the slice representation. Object automorphisms (e.g. the diabolical* $\mathbb{Z}/2$ *phase) are the* 2*-morphism data.*

Proof. A groupoid fibration presents a quotient stack; stratification of a quotient by stabilizer type and the étale slice model at a point are the content of the Luna slice theorem for reductive group actions. RG's strata and their $\mathrm{Sym}_0(m)$ slice representations are instances. □

Remark 13.2 (Worked instance: the diabolical phase). Deform $\mathcal{G}_0$ so that two null eigenvalues collide, i.e. approach a codimension-two conical intersection modelled locally by $A(\theta) = r\left(\begin{smallmatrix}\cos\theta & \sin\theta\\ \sin\theta & -\cos\theta\end{smallmatrix}\right)$, with $\mathrm{gap}/2r = 1$ at the cone (verified). Transporting the null frame once around the degeneracy returns it multiplied by -1 (holonomy overlap $= -1.000$, verified): the $\mathbb{Z}/2$ diabolical phase. In the stack of Theorem 13.1 this is the object automorphism at the double-eigenvalue stratum, whose slice representation is the weight-two $\mathrm{O}(2)$-action—exactly the 2-morphism datum. The phase is a standard stacky automorphism; it is not new.

Residual 13.3 (The bridges are not stacky consequences). Higher category theory organizes the moduli and its automorphisms but does not force the cross-realization bridges $g = \Sigma^{-1}$, $g_F = J^\top\Sigma^{-1}J$, and $\mathrm{I\!I} =$ connection mixing. Those identifications are structure added to the stack, not derived from it.

Verdict 13.4 (Attack X). Collapse of the moduli and stratification into stack theory. Residual: the cross-realization bridges.

14 Two further objects: reverse reading and reparametrization rigidity

RG carries two structures not yet attacked: the claim that the null sector can "read" the resolved sector symmetrically, and the claim that the resolved sector is rigid under reparametrization. We attack both, to check they are not hidden primitives.

Theorem 14.1 (Reverse reading is the symmetric Gaussian conditional)**.** *For jointly Gaussian (y_R, y_N) the reverse estimator is $\mathbb{E}[y_R \mid y_N] = C\Sigma_N^{-1}y_N$ with residual $\Sigma_R - C\Sigma_N^{-1}C^\top$, and the reverse canonical correlations—the singular values of $\Sigma_N^{-1/2}C^\top\Sigma_R^{-1/2}$—equal the forward ones. Hence "the hole reads the resolved sector" is not a new operation; it is the conditional expectation with the blocks exchanged, and it carries the same invariant.*

Proof. Exchange R and N in Theorem 4.3. The whitened cross-block $\Sigma_N^{-1/2}C^\top\Sigma_R^{-1/2} = (\Sigma_R^{-1/2}C\Sigma_N^{-1/2})^\top$ is the transpose of the forward K, and a matrix and its transpose have equal singular values; hence the canonical correlations coincide. The geometry is symmetric in $R \leftrightarrow N$; the observational asymmetry (we condition on R) is a choice, not a structure. □

Remark 14.2 (Worked instance on $\mathcal{G}_0$). The forward and reverse canonical correlations of $\mathcal{G}_0$ both equal $\{0.8205, 0.6262\}$ (verified). Reverse reading adds no invariant to the forward theory; it is a consistency test, and on $\mathcal{G}_0$ it passes identically.

Theorem 14.3 (Reparametrization rigidity is column-space invariance)**.** *For any invertible reparametrization $A \in \mathrm{GL}(\dim\Theta)$, $\mathrm{Im}(JA) = \mathrm{Im}(J)$; the resolved sector is invariant. This is the elementary invariance of the column space of a matrix under right multiplication by an invertible matrix, not a new rigidity phenomenon.*

Proof. $\mathrm{Im}(JA) = \{JAx : x\} = \{Jy : y\} = \mathrm{Im}(J)$ since A is a bijection. Equivalently the orthogonal projector onto the column space is unchanged; numerically, for $\mathcal{G}_0$ the two projectors agree to 8.1×10^{-16}. □

Verdict 14.4 (Further objects). Both collapse. Reverse reading is the transposed Gaussian conditional; rigidity is column-space invariance. Neither is a new primitive, and both reduce inside Attacks I and IV.

15 What each foundation forgets

Before the universality test we make explicit, for each framework, the forgetful functor it imposes: the structure it must drop to receive RG. This is the mechanism behind the gaps in the reduction table, and it shows the gaps are not accidents of presentation.

Proposition 15.1 (Forgetful profiles)**.** *Each framework X acts on the data $(J, g, \Sigma, atlas)$ of a resolution geometry by a forgetful functor Φ_X that retains exactly:*

(1) ***Linear algebra / operator theory*** *retains (J, g, Σ) in a single chart and forgets the atlas: no globalization, no monodromy.*

(2) ***Information geometry*** *retains the pullback $g_F = J^\top\Sigma^{-1}J$ and forgets the ambient blocks (Σ_R, C, Σ_N): no Schur floor, no conditional recovery (Theorem 7.3).*

(3) ***Differential geometry*** *retains $(g, \mathrm{I\!I}, \nabla^\perp)$ and forgets Σ: no covariance, no admissibility inequality.*

(4) ***Lie groupoid / sheaf*** *retains the atlas and its cocycle and forgets the fibre's internal geometry: no metric, no floor, no curvature.*

(5) ***Category theory*** *retains the exact sequence and transport and forgets the metric (Lemma 10.2): no orthogonal null, no correlations, no floor.*

(6) ***Homological algebra*** *retains the extension class and forgets the order and inversion structure (Theorem 11.1): no floor.*

No Φ_X is faithful on all of RG; each has a nontrivial kernel of forgotten structure.

Proof. Each clause is the residual established in the corresponding attack: (1) Residuals 4.8,5.5 plus the "—" globalization rows; (2) Theorem 7.3; (3) the residual after Theorem 8.3; (4) the residuals of Attacks VI and IX; (5) Lemma 10.2 and Residual 10.3; (6) Theorem 11.1. □

Corollary 15.2 (The forgotten structures do not overlap into a single survivor)**.** *The union of forgotten structures over all X covers every non-skeleton RG object: covariance blocks, orthogonal null, Schur floor, canonical correlations, second-order lift, and globalization are each forgotten by some framework. Hence no framework retains all of them, and—since the retained structures of different frameworks are genuinely different (metric vs. atlas vs. order)—no framework's retained profile contains another's entirely.*

Proof. Immediate from Proposition 15.1: each listed object appears in some clause's "forgets" list, and the retained lists are pairwise incomparable (each contains something another drops), so none dominates. □

16 Attack XI — The Universality Test

The first ten attacks each eliminated *some* RG objects. The decisive question is whether they can all be eliminated *at once*, inside a single framework.

Definition 16.1 (Reduction table)**.** For each RG object o and framework X, set $r(o, X) = 1$ if o reduces canonically into X under the Elimination Principle; 0 if it provably does not; and "—" if it is not expressible in X without importing another framework's data. Table 1 records the verdicts of Attacks I–X.

RG object	LinAlg/Op	InfoGeo	DiffGeo	Groupoid	Category	Homology
resolved/unresolved skeleton	1	1	1	1	1	1
transport	1	1	1	1	1	1
symmetry quotient / strata	1	1	1	1	1	1
globalization / monodromy	—	—	1	1	—	—
metric null (orthogonal)	1	1	1	—	$0^\dagger$	0
coupling (as extension)	1	1	1	1	1	1
Schur floor (admissibility)	1	0	—	—	$0^\dagger$	0
Gaussian recovery	1	0	—	—	$0^\dagger$	—
canonical correlations	1	1	—	—	$0^\dagger$	—
second-order lift $\mathrm{I\!I}$	—	—	1	—	—	—
Gauss–Codazzi–Ricci	—	—	1	—	—	—

Table 1: Canonical reducibility of RG objects by framework. "1" reduces; "0" provably does not; "—" not expressible without importing another framework's data. †: representable only after metric/order enrichment.

Theorem 16.2 (No single framework eliminates RG)**.** *There is no framework X among those tested with $r(o, X) = 1$ for every RG object o. Equivalently, there is no faithful functor F : $\mathrm{RG} \to X$ into a single foundation preserving all canonical reductions simultaneously.*

Proof. Every column of Table 1 contains an entry that is 0 or "—", and each such gap is a proven obstruction:

- **LinAlg/Op**: cannot express intrinsic globalization/monodromy or the second-order curvature constraints without bundle data (the "—" rows);
- **InfoGeo**: cannot force the covariance blocks (Theorem 7.3), hence not the Schur floor or Gaussian recovery (the 0 rows);
- **DiffGeo**: does not supply the covariance/statistical layer (residual after Theorem 8.3; the Schur/recovery rows are "—");
- **Groupoid/Sheaf**: globalize but do not supply the fibre (the metric, Schur, and curvature rows are "—");
- **Category**: represents the skeleton and coupling-as-extension but not the metric null, canonical correlations, or Schur floor without enrichment (Residual 10.3);
- **Homology**: provably cannot express the Schur floor (Theorem 11.1).

Since no column is all-ones, no single framework reduces every object; a faithful all-preserving functor would make some column all-ones, so none exists. ☐

Residual 16.3 (The obstruction is the residual). Ω_{RG} is precisely the obstruction of Theorem 16.2: the impossibility of eliminating RG within one foundation. It is not a new object inside any X; it is the pattern of Table 1—each object reducible somewhere, no framework everywhere.

Remark 16.4 (Why this is not a triviality). That a composite object uses several kinds of mathematics is by itself unremarkable; a Riemannian manifold with a measure is not captured by pure topology. What makes Theorem 16.2 content rather than truism is proved in §17 and formalized in §16.1: the several realizations are not independent decorations but the *same* object seen through different foundations, glued by canonical bridges. The non-triviality is the coherence.

16.1 The obstruction as a descent class

Theorem 16.2 is, so far, a reading of a table. We now give it the structural form that makes "no single functor" a genuine obstruction rather than an enumeration, by running one level up the same descent argument that Attack IX ran on the base.

Construction 16.5 (The site of foundations)**.** Let $\mathfrak{F} = \{X_1, \ldots, X_6\}$ be the frameworks of the columns of Table 1 (LinAlg/Op, InfoGeo, DiffGeo, Groupoid, Category, Homology), regarded as objects of a small category whose morphisms are the standard forgetful/enrichment functors between them (e.g. metric → linear, immersion → groupoid via holonomy, covariance → information via $g = \Sigma^{-1}$). For each X_i let $U_i \subseteq \mathcal{E}$ be the set of RG objects that reduce into X_i (the 1-entries of column i). The U_i form a cover of the object set $\mathcal{E}$: every RG object reduces somewhere (Complete Elimination Theorem, part (a)).

Lemma 16.6 (Overlaps carry the bridges)**.** *On each overlap $U_i \cap U_j$ the two reductions of a shared object are related by a canonical bridge functor $\phi_{ij} : X_i \supseteq U_i \cap U_j \to X_j$, given on the generators by the identities of Theorem 17.3: ϕ between covariance and information is*

$g = \Sigma^{-1}$; between immersion and gauge is Gauss–Weingarten; between covariance and moduli is the canonical-correlation invariant. These satisfy $\phi_{ii} = \mathrm{id}$ and $\phi_{ij}\phi_{jk} = \phi_{ik}$ wherever all three are defined.

Proof. Each ϕ_{ij} is the identification of the two framework-images of one RG object, which exists precisely on the overlap (where the object reduces both ways) and is the bridge of Theorem 17.3. The cocycle identities are the transitivity of these identifications: an object reduced three ways is one object, so the composite identification is the direct one. □

Theorem 16.7 (Ω_{RG} is a nonzero descent class)**.** *The reductions $\{U_i\}$ with the bridge cocycle $\{\phi_{ij}\}$ constitute descent data for RG on the site $\mathfrak{F}$. A single faithful functor $F : \mathrm{RG} \to X$ into one framework exists if and only if this descent datum is trivial, i.e. if some $U_i = \mathcal{E}$ (a global section on one chart). By Theorem 16.2 no $U_i = \mathcal{E}$; equivalently the cover $\{U_i\}$ admits no refinement to a single chart. Therefore the descent datum is nontrivial, and*

$$\Omega_{\mathrm{RG}} = [\{\phi_{ij}\}] \in \check{H}^1(\mathfrak{F}, \underline{\mathrm{Bridge}}) \setminus \{0\}$$

is a nonzero Čech class: RG is a nontrivial "bundle of foundations", glued by the bridge cocycle, with no global trivialization into a single foundation.

Proof. That $\{U_i\}$ covers $\mathcal{E}$ is part (a) of the Complete Elimination Theorem; Lemma 16.6 supplies a cocycle on overlaps. A global functor into X_i is a section over the single chart U_i, which requires $U_i = \mathcal{E}$; the cocycle is a coboundary iff such a global section exists on some chart. Since every column of Table 1 has a 0 or "—", no $U_i = \mathcal{E}$, so no coboundary trivialization exists, and the class is nonzero. This is the one-level-up image of the descent argument of Theorem 12.1. □

Remark 16.8 (What the descent class buys)*.* This upgrades the central claim from "the table has no full column" to "RG is a nontrivial torsor over the site of foundations, and Ω_{RG} is its gluing class." The residual is thereby exhibited as an actual cohomological obstruction of exactly the kind RG studies on its base—now applied to RG's own relationship to mathematics. RG is the geometry of partial observability, and its own irreducibility is a partial-observability phenomenon: no single foundation observes all of it.

17 Attack XII — The Irreducibility Test

Two questions remain: what survives *every* framework, and is the residual of §16 genuine content or an artifact of bolting parts together?

Theorem 17.1 (The derivation-stable core)**.** *The rows of Table 1 that are* 1 *in every column are exactly the resolved/unresolved decomposition (image and cokernel), transport between local identifications, the symmetry quotient with its strata, and—wherever a base with nontrivial loops is admitted—the globalization/monodromy obstruction. These form the geometry of partial observability and are forced by every observation-theoretic foundation.*

Proof. Kernel/image/cokernel and functorial transport occur in linear algebra, operator theory, information geometry, immersion geometry, category theory, and homology (Theorems 4.1–4.6, 5.1, 6.1, 7.1, 8.1, 10.1, 11.1); the symmetry quotient and strata occur in all via the quotient/slice construction (Theorem 13.1); monodromy is the shared globalization (Theorems 9.2, 12.1). Their intersection is the stated core. □

Remark 17.2 (The core is stable but not new)*.* Theorem 17.1 is a negative result for novelty at the level of primitives: the stable core is assembled entirely from established constructions. Partial observability, as a bare skeleton, is not a new mathematical object; if RG rested on the skeleton alone it would be eliminable by "ambient linear algebra plus a bundle."

Theorem 17.3 (The residual is coherence, not composition). *Ω_{RG} is not the trivial statement "RG uses several kinds of mathematics." It is the assertion, with proof, that the framework-specific realizations are the* same *object, glued by canonical bridges:*

(B1) $g = \Sigma^{-1} \Rightarrow g_F = J^\top \Sigma^{-1} J$ *(covariance $\leftrightarrow$ information);*

(B2) $g_F = \left(\Sigma_R - C\Sigma_N^{-1}C^\top\right)^{-1}$ *(Fisher = inverse Schur);*

(B3) $\mathrm{I\!I}$ *= off-diagonal block of the induced connection (immersion $\leftrightarrow$ gauge, Gauss–Weingarten);*

(B4) $\{\rho_i\}$ *= canonical correlations = complete gauge invariant on each stratum (covariance $\leftrightarrow$ moduli).*

Each is a theorem, not a choice; they are mutually consistent; and no single framework of §§4–13 forces the web.

Proof. (B1) is Theorem 7.2; (B2) is its inverse-Schur half, from the block-inverse formula. (B3) is Theorem 8.1. (B4): the canonical correlations are the singular values of $K = \Sigma_R^{-1/2} C \Sigma_N^{-1/2}$ (Proposition 4.5), and on a stratum with an eigenvalue of multiplicity m the residual gauge is $\mathrm{O}(m)$, under which a coupling block is classified completely by its singular values (a matrix modulo one-sided orthogonal action is determined by, and determines, its Gram matrix, by the singular value decomposition); by the Luna slice model (Theorem 13.1) these assemble to a complete invariant on every stratum. Consistency: the first-order condition (Schur floor, Theorem 4.1) and the second-order condition (Gauss–Codazzi–Ricci, Theorem 8.2) constrain disjoint data—Σ_N given (Σ_R, C), versus $(\mathrm{I\!I}, \nabla^\perp)$ given g—linked only through the Gauss equation; since $\mathrm{I\!I} = 0$ satisfies the second-order equations whenever the induced metric is flat, and a compatible $\mathrm{I\!I}$ exists otherwise by the existence half of Theorem 8.3, the two conditions are never over-determined. That the web is not forced by any single framework is Theorem 16.2. $\square$

18 The Complete Elimination Theorem

Theorem 18.1 (Complete Elimination Theorem). *Under the Elimination Principle applied through the twelve attacks:*

(a) (No surviving primitive.) *Every individual RG object reduces canonically to an established object in at least one framework; no RG object is an irreducible new primitive.*

(b) (No single foundation.) *No single framework reduces every RG object; there is no faithful all-preserving functor $F : \mathrm{RG} \to X$ (Theorem 16.2).*

(c) (Stable core.) *The structures reducible in every framework are exactly the geometry of partial observability—resolved/unresolved quotient, transport, symmetry quotient, monodromy (Theorem 17.1).*

(d) (Irreducible residual.) *Ω_{RG} is non-empty and equals the coherent web of canonical bridges identifying the covariance, information, immersion, and gauge realizations as one object over the stable core (Theorem 17.3); it is an irreducible* organization *of established structures, not a new primitive.*

Proof. (a) is the conjunction of the reduction theorems of Attacks I–X. (b) is Theorem 16.2. (c) is Theorem 17.1. (d) is Theorem 17.3 with Residual 16.3. $\square$

Corollary 18.2 (The dichotomy, resolved). *The dichotomy "either $\mathrm{RG} \simeq X_i$ for some established X_i, or there is a minimal obstruction Ω_{RG}" resolves in the second case: $\Omega_{\mathrm{RG}} \neq \varnothing$, and it is precisely the multi-foundational coherence, not a new object. RG is neither reducible to a single foundation nor a new branch disjoint from existing mathematics.*

19 What Resolution Geometry genuinely adds

At the level of primitives the elimination succeeds: nothing in RG is a new mathematical atom, and any claim of a previously unknown geometric primitive is refuted. At the level of the whole it fails, and the precise form of that failure is the contribution.

1. Partial observability as a geometry. The stable core (Theorem 17.1) treats finite resolution as a geometric object: missing directions (cokernel), their comparison (transport), their symmetry (quotient/strata), and their global obstruction (monodromy). This is not new mathematics but a new *subject*—the systematic geometry of what an observation cannot resolve—and it is derivation-stable across every foundation tested.

2. The bridge web. The genuinely irreducible content (Theorem 17.3) is the proven identification of four realizations as one object: the Mahalanobis bridge $g = \Sigma^{-1}$ aligning covariance with Fisher information; the inverse-Schur identity making the induced metric the Schur complement of the covariance; the Gauss–Weingarten identity making the second-order lift the sector-mixing block of the transport connection; and the canonical correlations serving at once as the coupling's information content and as the complete gauge invariant on every stratum. These are theorems binding foundations usually studied apart.

3. Consistency. The bridges are mutually compatible: first-order admissibility (Schur floor) and second-order integrability (Gauss–Codazzi–Ricci) constrain disjoint data and are not overdetermined. This is why the coherent object exists at all.

The unique portion, precisely. What RG uniquely is, after every canonical reduction, is the *coherent multi-foundational object of partial observability*: a single structure that is a covariance realization, a Fisher realization, an immersion realization, and a gauge realization at once, identified by canonical bridges and globalized by monodromy, no part of which is new but whose coherent assembly is forced by no single foundation. That is the irreducible remainder—an organization, not an atom, and real.

20 Non-claims

Non-claim 20.1. RG is not shown to be a new mathematical primitive. Attacks I–X reduce every RG object to established mathematics; any claim of a novel invariant unknown to existing theory is refuted by this paper.

Non-claim 20.2. Ω_{RG} is not a new object inside any framework. It is the obstruction to a single faithful functor (Theorem 16.2) and the coherence of the bridge web (Theorem 17.3); its novelty is organizational.

Non-claim 20.3. The stable core is not claimed to determine a physical system, nor to select a covariance realization uniquely. Covariance geometry is one strong realization, distinguished by computability (the Schur floor) and explicit recovery, not by derivational necessity.

Non-claim 20.4. No physical claim is made anywhere in this paper. The only physical identity in the wider program (fluctuation–dissipation) is inherited, equilibrium-bounded, and not invoked here.

21 Conclusion

We set out to eliminate resolution geometry, assuming a reader who knows all of modern mathematics and forbidding any new definition while a canonical reduction remained. The attempt succeeds against every RG object and fails against RG as a whole. Each primitive dissolves: resolved and unresolved are image and cokernel; the null sector is an orthogonal complement; transport is a connection or groupoid; the coupling is a block, an extension class, or a canonical correlation; recovery is least squares; recursion is recursive Schur; the second-order lift is the second fundamental form; globalization is monodromy and descent; the strata are a slice quotient. There is no surviving atom.

What survives is not an atom but an organization. No single foundation absorbs the whole, because the reductions land in different foundations, and the object RG names is the coherent identification of those foundations—covariance, information, immersion, gauge—as one structure over the geometry of partial observability, bound by canonical bridges and proven consistent. This is the residual: irreducible not because it is new mathematics, but because it is a single object that no single branch of mathematics contains.

The honest verdict is therefore neither triumph nor collapse. RG is not a new primitive; the elimination refutes that. RG is also not reducible to any one established theory; the universality test refutes that. RG is the multi-foundational geometry of incomplete observation: an assembly whose parts are old, whose bridges are theorems, and whose coherence is the contribution.

22 Precedents and disanalogies: is coherence itself new?

The claim of §19 must survive its strongest objection: mathematics already contains objects that are several established structures at once, glued by a compatibility law. If "coherent multi-structure" were itself the novelty, RG would have famous predecessors and no claim to originality of *type*. We confront this directly, because the honest value of the elimination depends on it.

Proposition 22.1 (Kähler precedent)**.** *A Kähler manifold is simultaneously a Riemannian manifold (M, g), an (integrable) complex manifold (M, I), and a symplectic manifold (M, ω), glued by the compatibility identities*

$$I^2 = -\,\mathrm{id}, \qquad \omega(X, Y) = g(IX, Y), \qquad \nabla I = 0.$$

This is a coherent multi-structure object: three established structures identified as one by canonical bridges, none of which alone is new [21].

Proposition 22.2 (Frobenius precedent)**.** *A Frobenius manifold carries a flat metric, a compatible commutative associative product on the tangent bundle, and an identity and Euler field, glued by the WDVV associativity equations [22]. Again: several established structures made coherent by a compatibility law.*

Corollary 22.3 (The pattern is not new)**.** *The abstract pattern "one object that is several established structures at once, glued by compatibility" is instantiated by Kähler and Frobenius geometry. Therefore RG cannot claim novelty merely for being a coherent multi-structure object; Theorem 17.3 is not, by itself, a claim of a new* kind *of mathematics.*

This is a genuine limitation and we state it plainly. What then, honestly, distinguishes RG from these precedents? Three things, none of which is "coherence in the abstract":

(i) **Foundational span.** Kähler and Frobenius geometry glue structures *within* differential geometry (metric, complex, symplectic; metric, product, grading). RG's bridges cross foundations usually kept apart: ordered covariance algebra (Schur floor), information geometry (Fisher), submanifold geometry ($\mathrm{I\!I}$, Gauss–Codazzi–Ricci), and gauge globalization (monodromy, character variety). The reduction table (Table 1) has no all-ones column precisely because its rows land in mutually distant foundations; the Kähler table would have an all-ones column (differential geometry).

(ii) **Subject.** The precedents are structures on a manifold chosen for its own sake. RG's object is forced by an external situation—finite resolution of an observation—so its "compatibility law" is not imposed for elegance but derived from the constraint that a covariance realization, a Fisher realization, and an immersion realization describe *the same partially observed system.*

(iii) **The floor is an inequality, not an identity.** Kähler/Frobenius compatibility are equalities ($\nabla I = 0$, WDVV). RG's first-order bridge includes an order relation, the Schur floor $\Sigma_N \succeq C^\top \Sigma_R^{-1} C$, which is why Attacks VII and VIII locate it outside categorical and homological reach. A coherence object whose gluing law is partly an inequality on an ambient cone is not the Kähler pattern.

Remark 22.4 (Honest downgrade). Corollary 22.3 downgrades the novelty from "a new kind of mathematics" to "a new instance, of a known kind, in a new subject, spanning an unusually wide foundational range, with a partly order-theoretic gluing law." We regard this as the correct, defensible statement of what RG is. It is less than a new branch and more than a repackaging; it is a Kähler-type coherence object for the geometry of partial observability.

A Explicit computations for the running example

All values quoted in Proposition 3.2 and the worked-instance remarks were computed from Construction 3.1. We record the derivations used.

Block-inverse identity (used in B2). For $\Sigma = \begin{pmatrix} \Sigma_R & C \\ C^\top & \Sigma_N \end{pmatrix} \succ 0$, the congruence of Theorem 4.1 gives $\Sigma = L^{-1} \begin{pmatrix} \Sigma_R & 0 \\ 0 & \Sigma_N - C^\top \Sigma_R^{-1} C \end{pmatrix} L^{-\top}$ with $L = \begin{pmatrix} I & 0 \\ -C^\top \Sigma_R^{-1} & I \end{pmatrix}$. Inverting,

$$\Sigma^{-1} = L^\top \begin{pmatrix} \Sigma_R^{-1} & 0 \\ 0 & (\Sigma_N - C^\top \Sigma_R^{-1} C)^{-1} \end{pmatrix} L.$$

Reading the upper-left block and applying the same identity with the roles of the two blocks exchanged gives $(\Sigma^{-1})_{RR} = (\Sigma_R - C\Sigma_N^{-1} C^\top)^{-1}$, which is invariant (d): $J^\top \Sigma^{-1} J = (\Sigma_R - C\Sigma_N^{-1} C^\top)^{-1}$ when J selects the resolved coordinates. Numerically the two sides agreed to 8.9×10^{-16}.

Canonical correlations and mutual information (used in c). With $K = \Sigma_R^{-1/2} C \Sigma_N^{-1/2}$, the singular values are $\rho = \{0.821, 0.626\}$. Then $-\frac{1}{2}\sum_i \log(1-\rho_i^2) = 0.8081$ and, independently, $\frac{1}{2}\log(\det \Sigma_R \det \Sigma_N / \det \Sigma) = 0.8081$; the two formulas agree to machine precision, confirming Proposition 4.5 on $\mathcal{G}_0$.

Schur floor (used in a). $\Sigma_N - C^\top \Sigma_R^{-1} C$ has eigenvalues $\{1.746, 3.684\} \succ 0$, and the full Σ has all positive eigenvalues, confirming the equivalence of Theorem 4.1 on $\mathcal{G}_0$.

Diabolical $\mathbb{Z}/2$ phase (used in Attack X). For $A(\theta) = r\begin{pmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{pmatrix}$ the eigenvalues are $\pm r$, so gap$/2r = 1$ at the cone. Continuously transporting the upper eigenvector around $\theta \in [0, 2\pi]$ returns it with inner product -1.000 against its start: a sign flip, the $\mathbb{Z}/2$ Berry phase. This realizes the weight-two slice representation of Theorem 13.1.

Monodromy (used in Attack VI). For structure group $\mathrm{O}(2)$ and monodromy $\rho(1) = R(2\pi/3)$, one has $\rho(1)^3 = \mathrm{id}$ and $\rho(1) \neq \mathrm{id}$; the flat structure is a nontrivial, non-identity-conjugate point of the character variety $\mathrm{Hom}(\mathbb{Z}, \mathrm{O}(2))/\mathrm{conj}$.

References

[1] P. R. Halmos, *Finite-Dimensional Vector Spaces*, Springer, 1974.

[2] F. Zhang (ed.), *The Schur Complement and Its Applications*, Springer, 2005.

[3] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.

[4] A. Ben-Israel and T. N. E. Greville, *Generalized Inverses: Theory and Applications*, 2nd ed., Springer, 2003.

[5] T. Kato, *Perturbation Theory for Linear Operators*, 2nd ed., Springer, 1995.

[6] M. Reed and B. Simon, *Methods of Modern Mathematical Physics I: Functional Analysis*, Academic Press, 1980.

[7] S. Amari and H. Nagaoka, *Methods of Information Geometry*, AMS/Oxford, 2000.

[8] T. M. Cover and J. A. Thomas, *Elements of Information Theory*, 2nd ed., Wiley, 2006.

[9] S. Kobayashi and K. Nomizu, *Foundations of Differential Geometry*, Vol. I, Wiley, 1963.

[10] S. Kobayashi and K. Nomizu, *Foundations of Differential Geometry*, Vol. II, Wiley, 1969.

[11] M. P. do Carmo, *Riemannian Geometry*, Birkhäuser, 1992.

[12] M. Spivak, *A Comprehensive Introduction to Differential Geometry*, Vol. IV, Publish or Perish, 1979.

[13] K. Tenenblat, "On isometric immersions of Riemannian manifolds," *Bol. Soc. Brasil. Mat.* 2 (1971), 23–36.

[14] W. M. Goldman, "The symplectic nature of fundamental groups of surfaces," *Adv. Math.* 54 (1984), 200–225.

[15] D. Husemoller, *Fibre Bundles*, 3rd ed., Springer, 1994.

[16] G. E. Bredon, *Topology and Geometry*, Springer, 1993.

[17] S. Mac Lane, *Categories for the Working Mathematician*, 2nd ed., Springer, 1998.

[18] C. A. Weibel, *An Introduction to Homological Algebra*, Cambridge University Press, 1994.

[19] D. Luna, "Slices étales," *Bull. Soc. Math. France, Mémoire* 33 (1973), 81–105.

[20] M. Brion, H. Kraft, and G. W. Schwarz, "Luna's Slice Theorem and applications," *Bull. Amer. Math. Soc.* 62 (2025), 643–694.

[21] A. Moroianu, *Lectures on Kähler Geometry*, London Math. Soc. Student Texts 69, Cambridge University Press, 2007.

[22] B. Dubrovin, "Geometry of 2D topological field theories," in *Integrable Systems and Quantum Groups*, Lecture Notes in Math. 1620, Springer, 1996, 120–348.